

Empowering Communities through IoT²: A Citizen Science Approach to Flood Monitoring and AI-supported Resource Optimization

Digital Resilience, Maker Education, and Energy-Aware IoT Development

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Abstract: This paper presents an integrated educational and technical framework for strengthening climate resilience and digital empowerment through citizen-driven environmental sensing and algorithmic literacy. The IoT²-Werkstatt⁴ is an open platform that enables students, teachers and communities to build and program IoT-based monitoring devices using open, visual and AI-supported tools. As a central use case, we examine a co-developed flood monitoring project carried out as part of the project “DigiSelfTrans”⁵. The platform fosters technical and civic empowerment in STEM education (Science, Technology, Engineering, Mathematics), particularly in rural areas. This paper outlines the conceptual foundations and implementation strategy of the IoT²-Werkstatt, presents selected outcomes (e.g. Energy savings for field devices), and discusses practical barriers such as system maintenance and equitable participation. It contributes to emerging hybrid models of sustainability education and participatory technology.

Keywords: Internet of Things and Thinking, Makerspace, Artificial Intelligence, Citizen Science, Climate Resilience, STEM Education, Energy Efficiency, Digital Inclusion

1 Introduction

Climate change and digital transformation are reshaping how education and public infrastructure must evolve. While institutions draft top-down policies, many rural communities lack the tools and capacity to act locally, resulting in growing gaps in participation. Rural regions often suffer from limited access to hands-on STEM (Science, Technology, Engineering, Mathematics) education [RM25], while also facing disproportionate exposure to climate-related threats such as flash floods or biodiversity loss. What is needed is an approach that enables especially young people to monitor,

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⁵ <https://www.umwelt-campus.de/iss/projekte/laufende-projekte/digiselftrans>

understand, and shape their environment through accessible, meaningful digital tools.

The IoT²-Werkstatt is an educational innovation that combines open hardware, visual programming, AI-assisted learning, and community-based sensing. Students, educators, and volunteers build real-world systems such as air quality indicators, irrigation tools, or flood monitors using plug-and-play sensors and microcontrollers. [GB23] A central example is a flood monitoring system from the DigiSelfTrans project, developed with schools, municipalities and the German Red Cross.

Rather than proposing a fixed solution, the IoT²-Werkstatt provides a learning ecosystem where people explore and iterate. This paper outlines the platform's structure, selected project results, and key open challenges such as device maintenance, sustained use, and data ethics—situating the work in broader debates around digital resilience and community empowerment. [Go24][Go19a][Go19b]

2 Informatics Education – From Algorithms to Applications

Digital education requires more than access to devices. It must foster algorithmic thinking and digital creation. The IoT²-Werkstatt introduces learners to coding through visual tools such as Ardublock.

Graphical programming allows learners to create sensor-based applications by connecting logic blocks - lowering entry barriers and removing syntax errors. Students work with variables, conditions, and loops, and later move on to finite state machines or event-based programming. A basic example might involve reading a CO₂ value and activating a traffic light system (green/yellow/red) depending on thresholds⁶.

Key informatics concepts include:

- Input–Process–Output logic,
- Data abstraction,
- Conditional execution and loops,
- Modularization and system decomposition,
- IoT²-Superblocks.

⁶ <https://www.heise.de/select/make/2020/5/2022015381334973804>

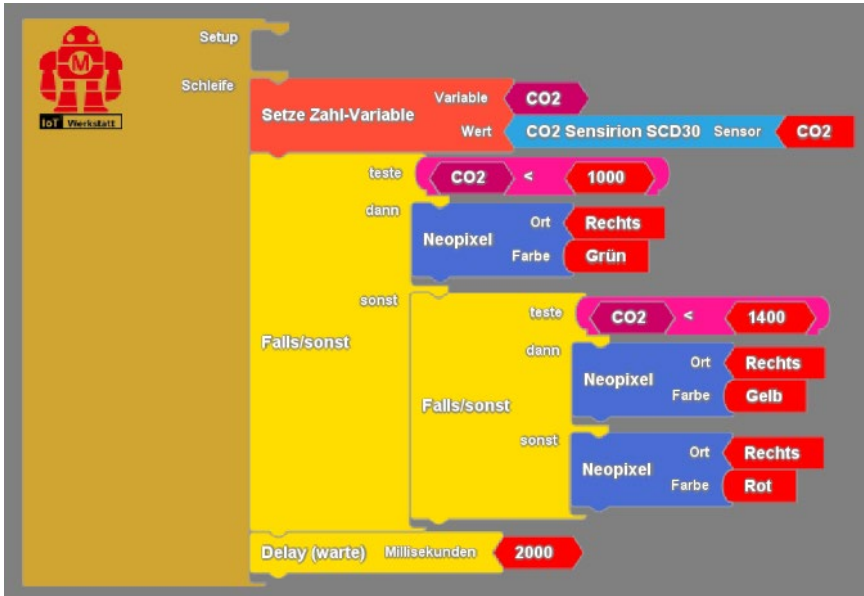


Figure 1: ArduBlock screenshot – simplified CO₂ traffic light logic

People are not only coding; they are building systems that respond to real environmental data. They measure, decide, and act - closing the loop between sensor data and tangible action. This promotes a systems-thinking mindset and strengthens problem-solving skills.

For those who wish to go further, the platform enables a smooth transition to text-based programming in Arduino C/C++. The visual blocks correspond directly to standard code structures, supporting progression from beginner to intermediate level.

In summary, the IoT²-Werkstatt offers an intuitive yet powerful entry point⁷ into informatics education, rooted in real-world relevance and designed for accessibility, creativity, and long-term learning.

3 Citizen Science in Practice: Flood Monitoring

As flash floods become more frequent and less predictable, decentralized and low-cost early-warning systems gain importance – particularly in regions where conventional infrastructure is either unavailable or underdeveloped. As part of the “DigiSelfTrans” project, the IoT²-Werkstatt has been set up to enable citizens to work together on developing and deploying flood sensors, among other things, under the motto

⁷ <https://www.umwelt-campus.de/iot-werkstatt/tutorials/schnellstart-octopus-anschiessen-und-einrichten>

“Understanding digitalization, promoting self-efficacy, mastering the climate crisis”. These robust, battery-powered devices are designed, assembled, and programmed in schools, communes and blue light associations, forming a grassroots infrastructure for local resilience.

Technical tasks include:

- Soldering components,
- 3D printing and laser cutting of building components,
- Programming energy-efficient DeepSleep cycles in Ardublock,
- Integration of Wi-Fi or LoRaWAN communication,
- Synchronizing with cloud platforms like ThingSpeak, Thingsboard or local MQTT brokers.

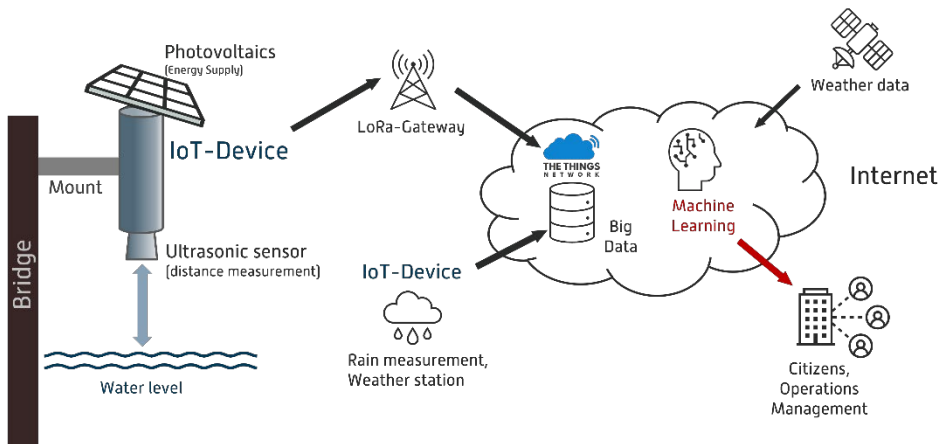


Figure 2: Functioning and networking of the flood gauge

The core device is a microcontroller with batteries/solar and an ultrasonic distance sensor, which we install in a PVC-tube on a bridge or above a small river. It measures the distance from the sensor to the water surface. Since we mount the sensor at a fixed height, a smaller distance means a higher water level - and in many locations, this directly correlates with potential flooding of nearby homes or infrastructure.

Once the measured water level rises above a critical threshold, the system triggers an early-warning notification. This information is used by citizens, schools, or first responder organizations such as the German Red Cross to take precautionary action. We are transmitting the sensor data via LoRaWAN, using a local LoRa gateway to reach cloud dashboards. There, we visualize water levels and compare them with predefined danger zones.

To enhance the reliability and relevance of this data, we intend to combine them with additional sources:

- Weather data, such as rainfall intensity and radar forecasts,
- Other IoT devices, such as humidity or soil sensors,
- Human observations and manual readings,
- And in the future, machine learning models for event prediction and pattern detection.

Together, this creates a participatory, multi-source warning system, empowering local actors not only to receive alerts, but to understand and act upon the situation. We can expand this form of digital resilience infrastructure modularly and adapt to different local contexts, especially in areas affected by past disasters such as the Ahr Valley, where citizens have an urgent need for transparent, local data to rebuild trust and reduce uncertainty during rainfall events. The process is guided by teachers, volunteers, and partners like the German Red Cross (DRK Kreisverband Vulkaneifel e.V.), making it a shared civic-tech endeavor.⁸



Figure 3: Dashboard prototype realized in Thingsboard⁹ for visualization and analysis of measurement data with water level thresholds for generating alarm systems

Each device can be programmed visually, including logic such as “transmit water level every 10 minutes unless battery < 3.4V” or “send alert if threshold X is crossed.” However, open questions remain. „How should data integrity be maintained if the device powers off or resets from Deep Sleep?”, “How can intermittent connectivity be handled, especially in

⁸ <https://messpegel.de/>

⁹ <https://thingsboard.io/>

remote areas?”, “What are the best physical mounting strategies, e.g., for attaching gauges to bridges or culverts?”, “Which communication technology (LoRaWAN vs. Wi-Fi) is most viable given local constraints?”.

Currently, deployments serve not only as learning environments, but also as testbeds for applied research. Feedback loops from real-world use help refine both software and hardware design-raising awareness for practical challenges like battery reliability, sensor calibration drift, and firmware robustness.

3.1 Digital Resilience as Public Infrastructure

The project expands the notion of disaster preparedness: resilience today includes sensor literacy, data fluency, and system thinking. Additional modules (e.g., air quality or humidity sensors) can transform school buildings into climate-aware learning spaces. By using Makerspaces as community labs, resilience-building becomes participatory, inclusive, and locally adaptable.

A key aspect, especially in communities affected by past disasters such as the Ahr Valley flood of 2021, is trust in data and local control. Many residents now experience existential anxiety during heavy rain events. It is therefore vital that they are not passive recipients of external alerts but can access and interpret environmental data themselves. The ability to install a sensor at the creek behind one’s home and check live data on a mobile dashboard creates not only technical transparency but psychological security and self-efficacy.

These infrastructures help build not only technical competencies but also social capital. Community workshops, student-led hackathons, and multi-generational repair events promote ownership, responsibility, and collaboration. Once implemented, such sensor networks can scale bottom-up, forming a low-cost, open-source backbone for environmental observatories and community alert systems.

3.2 Outlook: From Monitoring to Self-Empowered Resilience

The long-term vision goes beyond teaching how to build IoT-Systems. It seeks to enable citizens to identify risks of all kinds, deploy and maintain their own solutions, and manage local data autonomously. Upcoming developments include:

- Solder-free kits developed with *Make:magazine* and *Fab-Lab.eu*,
- Plug-and-play IoT² Dashboards for non-programmers,
- AI guidance tools for setup, learning and optimization,
- Modular extensions like rain-, soil moisture- or solar sensors.

Over time, the distinction between school project and civic infrastructure will fade. Each

IoT station becomes part of a living, learning network where environmental awareness, digital tools, and community resilience converge. Only speaking of the flood monitoring project, we built and installed more than fifteen flood gauges with schools. Until April 2026 there are at least six more devices planned with the optimized program code (5.1).

4 100 Ideas – A Platform for Creative Exploration

The true potential of the IoT²-Werkstatt lies not in any single use case, but in its ability to unlock creativity and technical exploration - especially among young learners. In hackathons, workshops and labs, hundreds of project ideas have emerged, often surprising educators with their originality and relevance. Unlike conventional development settings, these ideas are rarely overengineered or constrained by legacy thinking. Young people are less “locked in” to academic patterns and often propose intuitive, unexpected solutions to everyday problems. Examples¹⁰ include:

- A smart doorbell that notifies residents via smartphone,
- A connected pill box that reminds users and alerts their doctor on missed doses,
- An ice warning alarm that detects slippery conditions and triggers alerts,
- A DIY ECOTRON greenhouse that simulates climate conditions for plant research using cooling packs, fans, and cloud data input,
- A cocktail-mixing machine controlled by voice commands via Alexa and IoT protocols,
- A smart irrigation system using soil sensors and neural network logic for optimized water use.

These projects display the flexibility of the IoT² platform and its capacity to adapt to diverse learning goals and contexts. Each one is built with plug-and-play sensors and programmed via visual blocks by students as young as fourteen. Importantly, these learners also engage with questions of data ethics: Who owns the data? Should it be shared? What happens to personal or environmental information when it enters the cloud? Through APIs and dashboards, users begin to understand the social layer of digital systems and their role in shaping them. In this sense, the IoT²-Werkstatt is not defined by a specific project like flood monitoring. It is an open invitation to explore, design, and act on the world through code, sensors, and imagination.

¹⁰ <https://charta-digitale-vernetzung.de/deutschland-intelligent-vernetzt/div-konferenzen/iot-hackathon/>

5 Artificial Intelligence for Resource-Efficient IoT

The integration of AI into the IoT²-Werkstatt opens new possibilities - both in optimizing embedded systems and in supporting learners throughout the programming process. Generative AI tools such as ChatGPT are increasingly being used by students, teachers, and developers to reduce energy consumption, improve code quality, and democratize access to digital literacy.

5.1 Energy Optimization through AI

Microcontroller-based IoT devices are often constrained by limited power budgets, especially when battery- or solar-powered. In the DigiSelfTrans project, we used ChatGPT to analyse and refactor the monitoring-code with a specific goal: maximizing battery life. We present ChatGPT the C code generated by the code generator and ask for suggestions for optimization.¹¹

By prompting the AI to suggest optimizations, learners were introduced to techniques such as:

- DeepSleep modes that shut down the microcontroller between measurements,
- Dynamic sensor cycles that adjust based on weather conditions,
- Modular power management, e.g. disabling unused peripherals.

The result was a significant improvement in energy efficiency: Average current consumption dropped by 72%, from ~47 mA to ~13 mA.

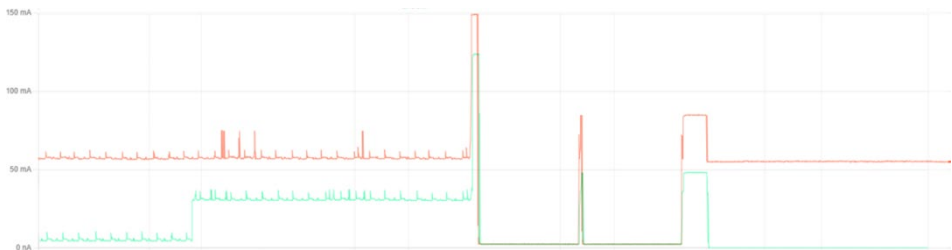


Figure 4: Energy usage overlay-comparison (duration 10s) of the flood gauge before (red line) and after (green line) AI-assisted optimization

This process not only made the systems more practical but also offered a real-world context for learning about software optimization, resource management, and AI-assisted development. The fact that we achieve these results through natural-language interaction with a chatbot demonstrates the accessibility and impact of AI tools in the maker context.

¹¹ <https://www.umwelt-campus.de/iss/projekte/laufende-projekte/digiselftrans/ki-assistent-ressourceneffizienz>

5.2 AI in Education and Resilience Empowerment

AI also transforms the way learners approach programming. Superblocks are a middle ground between HLL (High-Level Language - with guaranteed frustration) and Vibe Programming (without learning anything). The focus for the user is entirely on innovation and new ideas. At the same time, it trains algorithmic thinking. In the IoT²-Werkstatt, AI tools serve as digital tutors¹², helping students understand control structures, fix errors, and extend their own ideas. In natural language they can ask the system how to start with projects and ideas with basic questions like “How do I measure temperature every 10 minutes?”, “What does this condition mean?” or “How can I define thresholds for my sensor values?” These digital dialogues lower entry barriers by replacing code syntax with conversation. Learners are no longer afraid to make mistakes, and they gain confidence as they watch their questions transform into working code.

Teachers, in turn, use AI to generate custom worksheets and challenges, step-by-step video scripts, podcast outlines or differentiated tasks for students at varying skill levels. AI thus supports inclusive education: it adapts to the learner’s pace, language, and prior knowledge. It enables access to informatics even for those who struggle with abstract logic or complex syntax. By making computing conversational, it shifts the experience from memorization to exploration. AI in the IoT²-Werkstatt becomes more than a tool. It is a bridge between creativity and computation, supporting a new generation of learners in becoming not just consumers of technology, but co-creators of their digital future.

6 Conclusion and Outlook

The IoT²-Werkstatt links environmental monitoring, informatics education, and civic engagement. Participants build systems that reflect real-world complexity - gaining skills in problem-solving, systems thinking, and sustainability.

Since its origin at the 2016 IT Summit of the German Federal Government in Saarbrücken, with members of the IoT expert group, committed makers and the Environmental Campus of Trier University of Applied Sciences, the concept was evaluated in hackathons with over one hundred school teams. We integrate the initiative into curricula (e.g., in Rhineland-Palatinate), teacher training, and university courses like “Applied Computer Science & AI” at the Environmental Campus Birkenfeld.

Reliable figures are difficult because these are distributed DIY projects. By spring 2021, around three thousand CO₂ traffic lights were certainly in use in Germany alone (600 in classrooms, 2.000 Heise boards, plus many local projects); worldwide, a scale in the high four- to low five-digit range can be assumed.¹³ The IoT²-Werkstatt joined the MINT

¹²<https://chatgpt.com/g/g-67ff6858ea088191a94500af1066151c-iot-werkstatt-der-erste-mint-und-make-r-tutor>

¹³ <https://www.umwelt-campus.de/iot-werkstatt/news-detailansicht/5-jahre-diy-co2-ampeln>

Alliance of the BMFTR in 2021.¹⁴ Events such as hackathons, workshops and training are especially well received by young people and encourage further creative projects [DG17]. Ongoing dissemination happens through Maker Faire 2025 in Hannover, Make:magazine, and hands-on workshops inviting educators, volunteers, and civic actors to adopt and adapt the tools ensuring that the IoT²-Werkstatt continues to grow as a model for participatory digital resilience.

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¹⁴ <https://www.bildung-forschung.digital/digitalezukunft/de/bildung/iot>